

TULIHE, China

WMO: 504340

Lat: 50.483N

Lon: 121.683E

Elev: 2405

StdP: 13.46

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.0	-39.0	-48.5	0.3	-42.3	-44.4	0.5	-38.5	14.4	-5.0	12.5	-4.8	2.8	180	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	83.9	63.7	80.5	62.8	77.1	61.5	67.8	77.5	66.0	75.0	64.3	72.8	8.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	100.0	71.5	63.0	94.2	69.9	61.3	88.6	68.1	33.7	77.5	32.2	75.3	30.8	72.8	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.5	14.2	12.2	DB	-47.8	90.1	4.1	3.7	-50.7	92.7	-53.1	94.9	-55.4	97.0	-58.4	99.6	
			WB	-47.5	71.4	3.2	2.6	-49.8	73.3	-51.7	74.8	-53.5	76.2	-55.9	78.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.2	-18.5	-9.0	9.0	30.9	45.7	57.4	63.2	58.9	45.8	28.0	3.1	-14.1
	DBStd	29.99	8.30	9.72	11.88	8.98	7.21	6.33	4.67	5.93	7.70	9.48	11.85	9.38
	HDD50	10059	2123	1653	1270	574	171	13	1	7	171	683	1406	1987
	HDD65	14578	2588	2073	1735	1022	598	238	90	201	577	1148	1856	2452
	CDD50	1012	0	0	0	2	38	235	409	284	44	0	0	0
	CDD65	56	0	0	0	0	0	10	33	13	0	0	0	0
	CDH74	1421	0	0	0	1	79	426	575	299	41	0	0	0
	CDH80	332	0	0	0	0	18	111	135	62	6	0	0	0
	WSAvg	4.8	3.5	4.0	5.3	6.4	6.7	5.2	4.6	4.7	5.1	5.1	3.9	3.2
	PrecAvg	18.0	0.2	0.2	0.2	0.8	1.2	2.8	5.2	4.2	2.0	0.6	0.4	0.3
(15) Precipitation	PrecMax	24.5	0.4	0.4	0.6	1.3	2.4	6.2	9.2	9.3	4.4	1.3	0.6	0.7
	PrecMin	11.6	0.0	0.0	0.0	0.0	0.0	0.7	1.7	1.7	0.2	0.1	0.1	0.1
	PrecStd	3.3	0.1	0.1	0.1	0.3	0.7	1.4	1.8	1.9	1.1	0.4	0.1	0.2
	PrecStd	3.3	0.1	0.1	0.1	0.3	0.7	1.4	1.8	1.9	1.1	0.4	0.1	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	26.5	47.2	69.8	83.4	89.2	87.6	86.1	79.7	62.6	39.0	17.5
		MCWB	8.3	22.3	35.2	47.3	56.0	62.7	66.4	64.5	58.8	46.8	31.4	15.3
	2%	DB	5.1	20.3	40.8	61.9	75.4	83.1	83.9	81.0	73.2	55.2	31.6	9.9
		MCWB	3.7	17.1	31.4	43.2	53.0	61.7	65.7	64.2	56.1	42.4	27.3	8.4
	5%	DB	1.2	15.7	35.1	55.4	69.7	79.1	80.3	77.2	68.0	50.0	26.7	6.0
		MCWB	0.0	13.1	28.0	39.8	50.0	60.3	65.0	63.0	54.1	39.3	23.6	4.7
	10%	DB	-2.7	10.9	30.5	49.3	65.1	74.8	76.7	73.1	63.0	44.3	22.3	2.2
		MCWB	-3.8	8.8	25.0	36.8	47.9	58.7	63.9	61.5	51.4	36.1	19.9	1.2
	0.4%	WB	8.3	22.3	35.9	48.3	57.7	66.9	71.1	69.1	61.8	47.7	32.4	15.4
		MCDB	10.0	25.8	46.7	68.3	78.0	80.9	81.5	79.9	72.1	58.9	37.6	17.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	3.8	17.2	31.8	43.9	54.2	64.5	68.6	66.7	58.6	43.3	27.3	8.6
		MCDB	5.0	20.3	40.0	59.8	70.5	77.5	78.0	76.1	68.3	53.5	31.4	9.9
	5%	WB	0.1	13.1	28.3	40.6	51.9	62.5	67.1	64.8	55.8	40.2	23.8	4.8
		MCDB	1.2	15.7	34.9	54.0	66.4	73.5	75.8	73.0	65.2	48.8	26.7	5.9
	10%	WB	-3.6	8.9	25.1	37.6	49.3	60.6	65.4	63.0	52.9	36.9	19.9	1.2
		MCDB	-2.7	11.0	30.3	48.5	62.4	71.3	73.4	70.6	61.2	43.7	22.2	2.1
(35) Mean Daily Temperature Range	5% DB	MDBR	27.8	33.6	32.9	27.5	29.7	28.6	23.8	25.3	28.7	25.7	26.0	25.7
		MCDBR	28.8	34.9	32.6	38.7	40.6	38.0	31.9	32.7	37.6	35.2	28.6	25.5
		MCWBR	27.5	31.6	24.0	22.2	20.4	18.0	15.7	17.1	21.1	22.6	24.0	24.2
	5% WB	MCDBR	28.6	34.7	31.8	35.0	34.3	30.5	24.7	25.7	31.0	31.9	27.5	25.5
		MCWBR	27.4	31.5	23.6	20.5	18.3	16.1	13.6	14.5	17.8	21.0	23.1	24.3
(40) Clear-Sky Solar Irradiance	taub		0.252	0.277	0.327	0.420	0.489	0.462	0.430	0.397	0.350	0.323	0.281	0.262
	taud		2.182	2.135	2.050	1.988	1.897	2.084	2.242	2.311	2.386	2.346	2.256	2.127
	Ebn at Noon		247	270	273	258	245	254	261	265	267	251	233	219
	Edh at Noon		23	32	42	51	59	49	41	37	30	26	22	21
	RadAvg		510	884	1366	1640	1758	1871	1696	1546	1231	830	508	386
(45) All-Sky Solar Radiation	RadStd		19	35	78	99	156	176	141	152	99	63	30	26

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	N/A	N/A	-3.91	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	DBAvg	N/A	N/A	-3.62	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page